

Team Name: CImEx Compliance

Team Size: 4

Problem Statement:

From 2026 the EU charges for the carbon in imported steel goods. Indian exporters without verified data get priced on penalty defaults, and most cannot produce that data from the paperwork they have.

Problem Statement

The challenge or problem identified by the team.

- **Why now.** The EU carbon border tax (CBAM) became a real cost in 2026. The payable share rises from **2.5% in 2026 to 100% in 2034**. The first annual declaration is due **30 Sep 2027**.
- **The penalty.** Suppliers without verified data are priced on EU default values plus a **10% to 30% mark-up**. Indian goods become more expensive than rivals with data.
- **The scale.** India sold **€5.30 bn** of CBAM-covered goods to the EU in 2025.[1]
- **Why it is hard.** The data sits in scanned mill certificates, invoices and fuel bills. A verifier checks it line by line, and experts who can prepare it are scarce and costly. Most of a fastener's number is its steel, and only the mill has that data.
- **Who feels it.** Mid-size Indian metal exporters: fasteners, wire and pipes. We start in Chennai and Hyderabad.

1. Eurostat Comext, via CBAMGuide.

Proposed Solution

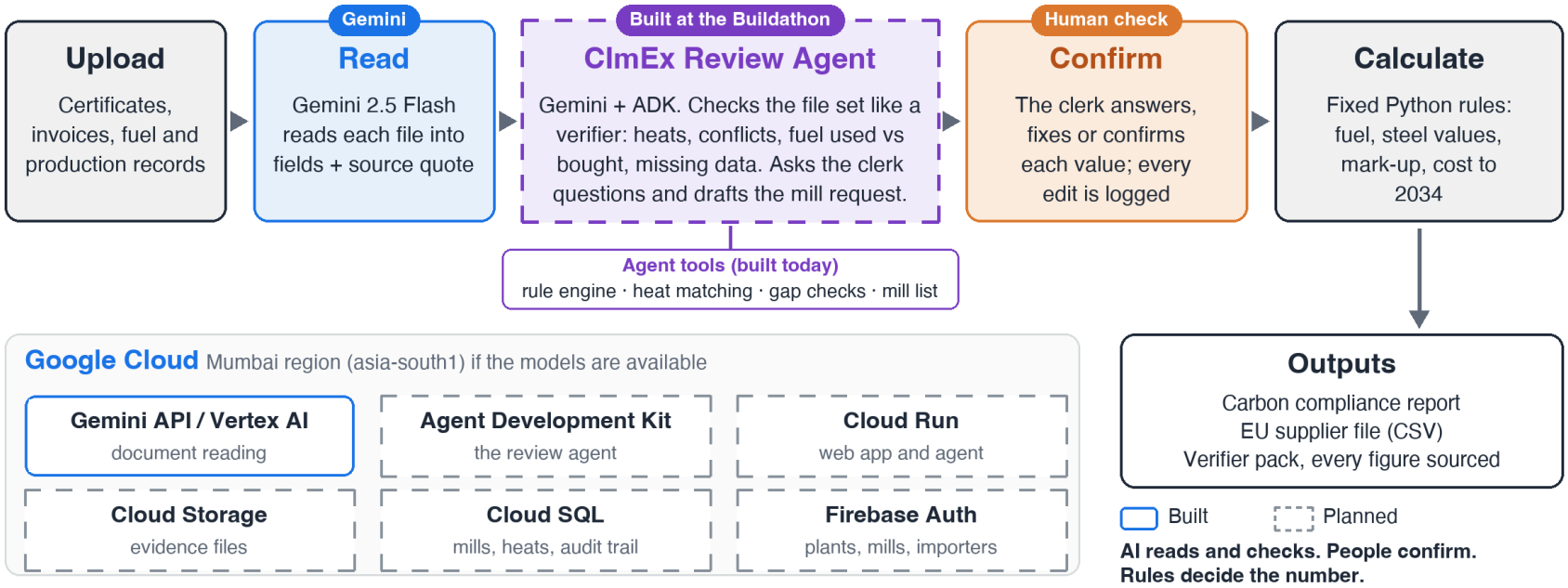
The idea and how it addresses the identified problem.

CImEx Compliance gives every exporter an AI compliance agent that prepares carbon data the way an expert would, and that an EU verifier can check.

- **Gemini reads.** The clerk uploads certificates, invoices, fuel and production records. Gemini turns each file into fields, each with its source line.
- **The CImEx Review Agent checks.** Built with Gemini and Google's Agent Development Kit, it reviews the whole file set like a verifier: heats without tonnes, conflicting documents, fuel bought but not logged, missing records. It asks the clerk short questions and drafts the data request to the mill.
- **A person confirms.** The clerk answers, fixes and confirms. Every edit is logged.
- **Rules decide the number.** The agent calls our rule engine as a tool. It can never change a figure itself, so the same input always gives the same result.
- **Network effect.** Verify a mill once, and every plant that buys its steel moves off the penalty default. In our demo, one mill sharing data takes a plant from **4.40 to 2.25 tCO2 per tonne**.

Solution Architecture

Architecture of the proposed solution



Google AI Tools

Google AI Studio, Google Antigravity and Google Agent Skills (used or planned to be used).

- **Google Agent Development Kit and Agent Skills** **Centre of our Buildathon build**
  The CImEx Review Agent. Skills for CBAM rules, heat matching, fuel checks and mill requests. Tools that call our existing rule engine and gap checks.
- **Gemini API** **Used**
  Reads scanned certificates, invoices and fuel bills into fixed fields, with the source quote for each value. It will also drive the agent's reasoning.
- **Google Antigravity** **Planned**
  Build and test the agent and the hosted app on Cloud Run during the Buildathon.
- **Google AI Studio** **Planned**
  Tune the reading prompts and output schemas on real certificate layouts, and measure accuracy per field.

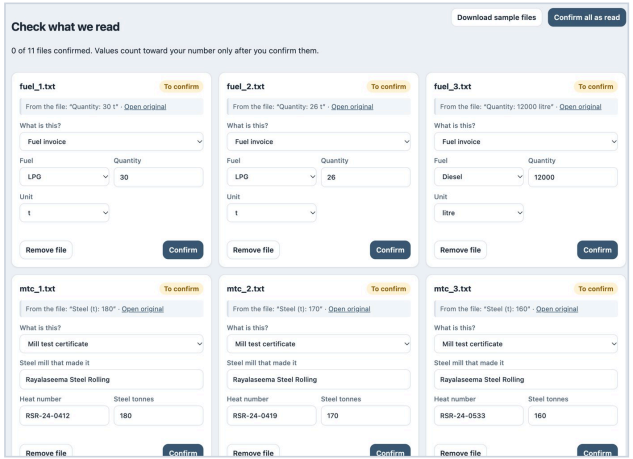
Google AI Models

Google AI model(s) used or planned for the proposed solution.

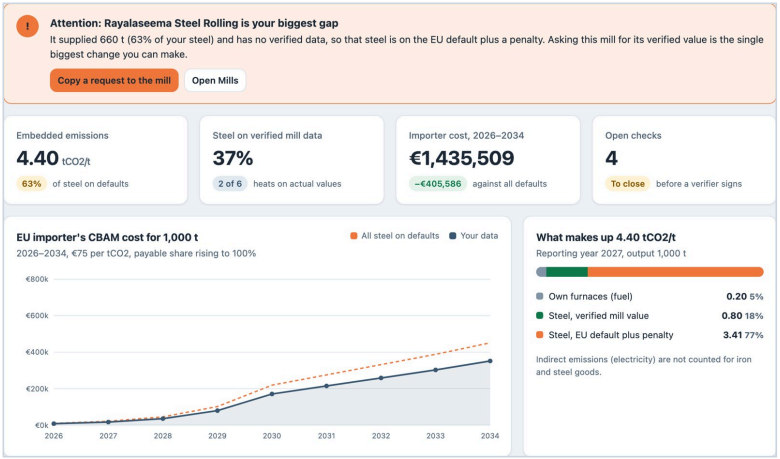
- **Gemini 2.5 Flash** **Used for reading**
  Multimodal and low cost per page. It is the model set in our code. It must read scans, stamps, phone photos and Indian number formats such as 1,00,000 kg.
- **Gemini for the agent** **Planned**
  The review agent reasons over the whole file set and plans its checks. We will test Gemini 2.5 Flash against Gemini 2.5 Pro for this and pick the one that finds the most real gaps.
- **Structured output.** Gemini returns fixed fields with the source quote. Our rules reject impossible values, such as negative tonnes.
- **Measured accuracy.** Every clerk correction is logged, so real documents give a field-by-field accuracy score for Gemini.
- **Vertex AI** **Planned**
  Run both on Google Cloud, in the Mumbai region if the models are available there.

Built today, and what we build at the Buildathon

https://hemanthdadi1.github.io/AI-Export-Compliance/



Live: each value with its source line, to confirm or fix



Live: 4.40 tCO2/t, the biggest gap and 4 open checks

Built and tested: Gemini reading, review and confirm, heat matching, gap checks, rule engine, report, verifier pack. Sample data is made up. **Buildathon build:** the CImEx Review Agent on ADK, hosted on Cloud Run. **Demo goal:** on the sample plant, the agent finds the 4 open checks, asks the clerk only what it cannot resolve, and drafts the request to the mill. The clerk approves and the report unlocks.

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Startup Buildathon

— India 2026

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Thank you

